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National Football League concussion lawsuit: what it means for other sports and observations from Singapore Rugby

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With the upcoming fifth International Conference on Concussion in Sport taking place in October 2016, the management of concussion in athletes will again be under increased scrutiny. However, it has rarely left the spotlight with 58 articles published in *BJSM* alone between 2012 and 2016!

LEGAL PRECEDENT

An off-field event fuelling this continued interest has been the American NFL players' lawsuit; a group of retired NFL athletes won a US$900 million out-of-court settlement from the organisation, stating they were not warned about brain injury risk and that cases were hidden from them. Upheld by the court of appeals, there are calls to include players with chronic traumatic encephalopathy in this decision. This may set a new legal precedent, and already another lawsuit has been filed by American footballers against the National Collegiate Athletic Association on similar grounds.

MEDIA INFLUENCE

Media attention on the concussion lawsuit in America highlights the potential medicolegal sequelae if concussion is (or is perceived to be) inadequately detected or managed. Commentary by medical and non-medical experts on mainstream and social media,[1] with analysis of medical decisions at sporting events such as the 2014 FIFA World Cup, often make health professionals responsible for the sideline assessment of concussion uneasy. Indeed, at the recent HSBC Rugby Sevens tournament in Singapore, careful assessment was conducted before individuals with suspected concussion were removed from competition; a decision that can alter the balance of play.

RUGBY'S REGULATED APPROACH

Pfister *et al*[2] reported that youth concussion rates (per 1000 exposure hours) were highest in rugby (4.18), ice hockey (1.20) and American football (0.53). Given that helmets do not significantly impact the risk of concussion in sport and the impact forces in rugby and American football are similar despite different tackling techniques,[3–5] it is surprising the incidence of concussion in each sport among youth athletes is not more similar.

World Rugby (WR) promotes proactive recognition and management of concussion, including regulatory changes, such as head injury substitutions which allow greater ease of application of the WR Head Injury Assessment (HIA) tool. Importantly, WR requires standardised training for all pitchside medical professionals, with concussion recognition/management being key components. Standardised evaluation and management protocols are now emphasised across the NFL, perhaps a vital step that needs to be replicated across all sports.

IS SOCCER LAGGING OR LEADING?

As a primary sponsor of all five International Consensus Conference(s) on concussion, clearly FIFA as an organisation supports improved concussion awareness, recognition and management. However, the broad reach of soccer as the world's most popular sport, from youth to the professional level, and in countries with variable resources and training of health professionals for the care of athletes, FIFA faces practical challenges of implementing a universal standard for the assessment and management of concussion. Thus, some medical practitioners may remain vulnerable when managing this condition.

SINGAPORE RUGBY OBSERVATIONS

In addition to the common challenges of explaining the importance of concussion to coaches, parents and athletes, a key challenge in Singapore has been the HIA system becoming 'lost in translation'. With its ethnic diversity and multiple home languages, lack of HIA translations has proved a challenge when hosting local and international events. Often, HIAs are conducted through non-medical translators who may not convey the seriousness of its assessment or the significance of positive findings.

MOVING FORWARD

With the constant spotlight on concussion, it is hoped that each point will be discussed at the 2016 conference. Although the NFL lawsuit represents the first successful litigation against a sporting league based on concussion management, it highlights the vulnerability of sporting organisations, especially those without adequate clinical guidance, education and policy adaptations to manage this condition. It is certainly possible that other sporting bodies will face litigation in the future, if suitable steps are not taken. Recent guidance for youth soccer concussion in the USA is a positive step,[7] but among pitchside medical professionals, there remains apprehension about being overly cautious versus underdiagnosing the problem.

A combination of legislative change and focused education of key stakeholders, including coaching staff, club management, families, athletes and medical teams, remains central to improving outcomes.[8] When a diagnosis of concussion is made, it is vital that we as healthcare professionals have the correct skills and expertise to manage it. Common criticisms of medical practitioners include using outdated resources and referring to non-specialists for further management. Often this is a reflection of variable education and exposure; those working in contact sport are more likely to keep updated, yet individuals from a non-medical or non-sports background may not have a similar skill set.

Some sports, including the NFL, have made significant progress since the 2012 International Conference on concussion in sport. One hopes the upcoming conference will address existing deficiencies and challenges, to further improve athletes welfare with evidence-based, best-practice guidelines for all sports.

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